

IT5002 Computer Systems and Applications Tutorial 9

Question 1.

Consider the following segment table:

Segment	Base	Length
0	219	600
1	2300	14
2	1327	580
3	1952	96

What are the physical addresses for the following logical addresses?

- (a) (0,430)
- (b) (1,10)
- (c) (2,500)
- (d) (3,400)

Which of these addresses, if accessed, would result in a segmentation violation?

Question 2.

Assume there is a 1,024KB segment where memory is allocated using the buddy system. Describe the configuration of the system as the following allocation requests are served:

- (a) Request 240 bytes
- (b) Request 120 bytes
- (c) Request 60 bytes
- (d) Request 130 bytes

Next, explain the configuration of the system as the blocks allocated first, third, and fourth are released.

Question 3.

This question is based on a "simple" file system built from the various components discussed in the lectures.

Partition Information (Free Space Information)

- Bitmap is used, with a total of 32 file data blocks (1 = Free, 0 = Occupied):

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	1	0	0	1	0	0	0	0	1	0	1
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	0	1	0	1	0	1	0	1	1	1	0	0	1	0	0

Directory Structure + File Information:

- Directory structures are stored in 4 "directory" blocks. Directory entries (both files and subdirectories) of a directory are stored in a single directory block.
- Directory entry:
 - o **For File:** Indicates the first and last data block number.
 - o **For Subdirectory:** Indicates the directory structure block number that contains the subdirectory's directory entries.
 - o The "/" root directory has the directory block number 0.

0			1			2			3		
y	Dir	3	g	File	0 31	k	File	6 6	i	File	1 3
f	File	12 2	z	Dir	2				h	File	27 28
x	Dir	1									

File Data:

- Linked list allocation is used. The first value in the data block is the "next" block pointer, with "-1" to indicate the end of data block.
- Each data block is 1 KB. For simplicity, we show only a couple of letters/numbers in each block.

0	1	2	3	4	5	6	7
11 AL	9 TH	-1 S!	-1 ND	23 GS	-1 SO	-1 :)	10 TE
8	9	10	11	12	13	14	15
31 RE	3 EE	28 M:	31 OH	19 SE	13 AH	4 IN	17 NO
16	17	18	19	20	21	22	23
30 YE	2 OU	1 ON	17 RI	26 EV	14 AT	21 DA	7 YS
24	25	26	27	28	29	30	31
-1 HO	18 ME	0 AL	30 OP	-1 -(	5 LO	21 ER	-1 A!

- (Basic Info) Give:
 - The current free capacity of the disk.
 - The current user view of the directory structure.
- (File Paths) Walkthrough the file path checking for:
 - **"/y/i"**
 - **"/x/z/i"**
- (File access) Access the entire content for the following files:
 - **"/x/z/k"**
 - **"/y/h"**

- d. (Create file) Add a new file `"/y/n"` with 5 blocks of content. You can assume we always use the free block with the smallest block number. Indicate **all changes required to add the file**.

Question 4.

Consider the following virtual page references:

1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6

How many page faults would occur for the following replacement algorithms, assuming three, four, and seven physical pages? Remember that physical pages are initially empty, so your first unique pages will cost one fault each.

- (a) LRU replacement
- (b) FIFO replacement

Question 5.

In this question we will consider a virtual memory system with 64 bytes per page, 10 bit virtual addresses and 9 bit physical addresses.

- a. What is the maximum size in bytes of the virtual memory? Physical memory?
- b. What is the maximum number of virtual and physical pages?
- c. Using the page table below, translate the following virtual addresses to physical addresses:

0, 15, 576, 987, 1020

	V	PPN
0	1	2
1	1	3
2	0	(13,5,17)
3	0	(12,5,6)
4	1	4
5	1	7
6	0	(10,3,1)
7	0	(3,4,1)
8	0	(7,5,5)
9	1	6
10	0	(6,3,13)
11	0	(7,1,16)
12	1	5
13	0	(6,4,1)
14	0	(6,8,3)
15	1	1